MASUMI BHATT

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EDUCATION

Northeastern University (Boston, MA) **September 2023 – August 2025**

Master of Health Informatics GPA: 3.95/4.0

Coursework: Introduction to Data Mining and Machine Learning, Data Handling and Retrieval, Health Data Analytics, Project Management

L.M. College of Pharmacy (Ahmedabad, India) **August 2017- June 2023**

Doctor of Pharmacy (PharmD) GPA: 3.8/4.0

Coursework: Biostatistics and Research Methodology, Pharmacotherapeutics, Pharmacology, Pharmacokinetics and Biopharmaceutics

SKILLS

Technical Skills: Python, SQL, R, SPSS, SAS, Tableau, Power BI, Amazon QuickSight, Snowflake, MongoDB, MS Excel

Analytics & ML: Statistical Modeling, Predictive Analytics, Regression, Machine Learning, ETL, Data Visualization, A/B testing

Healthcare & Cloud: HIPAA, HL7, FHIR, SNOMED CT, ICD-10, EPIC, Azure, AWS

PROFESSIONAL EXPERIENCE

Mass General Brigham Health Plan (Boston, MA) **January 2025– May 2025**

Data Analyst Co-op

- Automated **Healthcare Effectiveness Data and Information Set (HEDIS)** quality checks using **SQL scripts**, increasing audit success rates by 95% and improving data validation speed.
- **Collaborated with data engineering and clinical teams** to identify and resolve data inconsistencies, resulting in improved HEDIS measure reporting accuracy.
- Engineered SQL scripts to streamline the extraction, cleaning, and analysis of HEDIS data, reducing **processing time by 40%.**
- Automated manual reporting tasks through custom **SQL queries and Excel macros**, cutting down **reporting time by 30%** and increasing submission efficiency.
- Designed and maintained interactive **Tableau/Power BI dashboards**, enabling leadership to make data-driven decisions and improve care quality outcomes.

Sterling Hospital (India) **June 2022 – June 2023**

Clinical Data Analyst

- Implemented **Machine learning** models in **Python** to flag high-risk prescriptions, reducing adverse drug events by 77%.
- Evaluated 145 **Drug Interactions** using R, identifying patterns that led to updated clinical protocols and improved patient safety.
- Analyzed Medication Adherence patterns in 10,000+ patient records using statistical analysis, Tableau, and R, improving outcomes.
- Utilized **advanced analytics tools (R, SQL, Excel)** to monitor prescribing trends and clinical outcomes, optimizing hospital protocols, and ensuring better treatment efficacy.
- **Created data-driven reports** to track and assess the impact of hospital quality initiatives, presenting findings to senior management to guide hospital-wide improvements in care delivery.

L.M. College of Pharmacy (India) **August 2021 - May 2022**

Research Assistant

- Analyzed **MDR1 gene** data using R and SPSS, identifying genotype frequencies with 95% accuracy for pharmacogenomic insights.
- Applied **Predictive Modeling** to assess genetic variation's impact on drug metabolism and treatment outcomes.
- Processed genetic data from DNA extraction, PCR, and gel electrophoresis, ensuring accuracy for clinical analysis.

Central United Hospital **May 2021 - July 2021**

Data Analyst Intern

- Analyzed **large clinical datasets** (over 10,000 records) using Excel, Snowflake, and SQL, ensuring 98% data accuracy for insights.
- Utilized **Python and Tableau** to perform exploratory data analysis, build visualizations, and support data-driven decision-making.
- Ensured compliance with healthcare standards (**HIPAA, HL7**) while analyzing clinical data and building visualizations.

PROJECTS

Impact on Mental Health of Pregnant Women During COVID-19

- Analyzed a dataset of 10,000+ records to assess the effects of mental health and socioeconomic factors on birth outcomes.
- Applied **machine learning techniques** (e.g., Random Forest, SVM) to analyze the impact of anxiety and stress on birth outcomes.
- Created visualizations and trend analyses using Python and Tableau, improving **healthcare policy recommendations by 20%.**

Improving Heart Disease Prediction: Insight from Machine Learning Models

- Analyzed a heart disease dataset (1,000+ records) to identify **risk factors** and **enhance early detection**.
- Developed Logistic Regression, KNN, and Random Forest models, achieving **92.9% accuracy with Random Forest.**

Operative Procedure Integration Database

- Designed and implemented a **SQL database** to streamline **operative procedure data retrieval** for healthcare providers.
- Optimized queries, **reducing execution time by 30%,** and enhanced reliability to support better clinical decisions.

PUBLICATIONS

- Assessment of potential drug–drug interactions among outpatients in a tertiary care hospital: focusing on the role of P-glycoprotein and CYP3A4 (retrospective observational study).